

TDE Dataset Summary Statistics

Metric	Value	Notes
Number of Specimens	8	All archosaurs, ESRF Paleo CT
Projection Images (total)	240	8 specimens $\times$ 30
Projections per Specimen	30	6 ortho + 24 rotational
Orthographic Views	6 (3 axes $\times$ 2 directions)	Coronal, sagittal, transverse + flips
Rotational Views	24 (0°–345°, Δ 15°)	2D-after-MIP (sagittal MIP rotated)
Rotation Step	15°	Covers full 360°
Synthetic Images (total)	960	240 inputs $\times$ 4 deformations
Deformation Types	4	Compression, shearing, stretching, dissolution
Images per Deformation	240	Balanced; seed=42
Binary Masks (total)	960	One mask per synthetic image
Image Resolution	224 $\times$ 224 px	INTER_AREA resize from raw MIP
Image Format	PNG (uint8 grayscale)	Grayscale, no color channels
Random Seed	42	Reproducible with np.random.default_rng(42)
Label Format	CSV (11 columns)	image_id, specimen, deformation_type, params, ...